

0	4	.	1
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Formation of an enzyme-substrate complex increases the rate of reaction.

Explain how.

[2 marks]

0	4	.	2
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A scientist measured the rate of removal of amino acids from a polypeptide with and without an enzyme present. With the enzyme present, 578 amino acids were released per second. Without the enzyme, 3.0×10^{-9} amino acids were released per second.

Calculate by how many times the rate of reaction is greater with the enzyme present. Give your answer in standard form.

[2 marks]

Answer = _____ times faster

Turn over ►

Another scientist investigated an enzyme that catalyses the following reaction.



The scientists set up two experiments, **C** and **L**.

Experiment **C** used

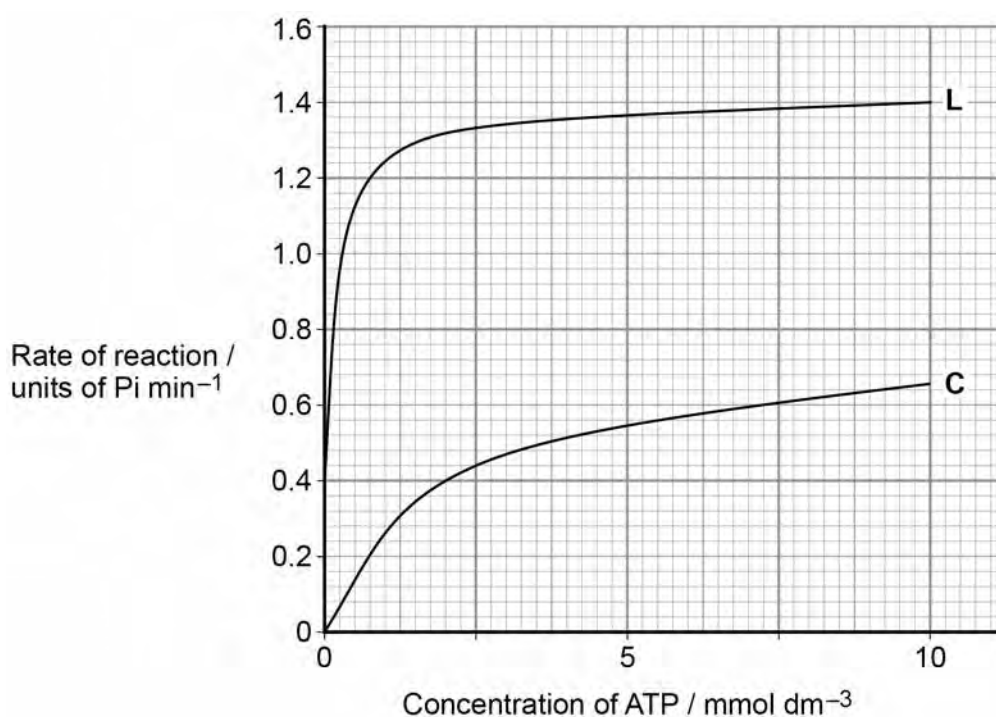
- the enzyme
- different concentrations of ATP.

Experiment **L** used

- the enzyme
- different concentrations of ATP
- a sugar called lyxose.

The scientists measured the rate of reaction in each experiment. Their results are shown in **Figure 5**.

Figure 5



0	4	.	3
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Calculate the rate of reaction of the enzyme activity with no lyxose at 2.5 mmol dm^{-3} of ATP as a percentage of the maximum rate shown with lyxose.

[2 marks]

Answer = _____ %

Question 4 continues on the next page

0 4 . 4

Lyxose binds to the enzyme.

Suggest a reason for the difference in the results shown in **Figure 5** with and without lyxose.

[3 marks]

[Extra space]

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